

Estimations

- Assumptions
 - Registered users: 400M
 - Daily Active Users (DAU): 120M
 - Peak concurrent streams: 20M
 - Average bitrate: 4 Mbps
 - Catalog: 80,000 titles, 800 added per day
- Bandwidth
 - $20M \times 4 \text{ Mbps} = 80 \text{ Tbps}$ at peak
 - With 2x headroom = $\sim 160 \text{ Tbps}$
 - Therefore video must be served from a distributed CDN, not the origin
- Storage
 - 8 encoded versions per title, $\sim 4 \text{ GB}$ each = 32 GB per title
 - $80,000 \times 32 \text{ GB} = \sim 2.5 \text{ PB}$
 - With replicas: $\sim 8 \text{ PB}$
 - Growth: $800 \text{ titles/day} \times 32 \text{ GB} = \sim 25 \text{ TB/day}$, $\sim 9 \text{ PB/year}$
- Requests
 - ~ 20 requests per viewing session
 - $120M \text{ sessions} \times 20 = 2.4 \text{ billion requests/day}$
 - $2.4B / 86,400 = \sim 28,000 \text{ RPS}$ average
 - Peak at 3x = $\sim 85,000 \text{ RPS}$
- Cache
 - Hot metadata, home rows, search results, sessions
 - $\sim 150 \text{ GB}$ per region
- Servers
 - 200 RPS per server
 - $85,000 / 200 = \sim 425 \text{ servers}$
 - x2 for redundancy across regions = $\sim 850 \text{ servers}$

Note: these cover the application tier only. CDN capacity for video delivery is separate and much larger.